Technical Challenge: Merchant Risk Scoring & Decisioning

Context

Clip is a fintech that processes card payments for merchants, both in-person (Card Present) and online (Card Not Present). Clip bundles many merchants together into "affiliations" that are sent to card issuers for approval; issuers judge Clip's overall trustworthiness based on the quality of these bundled transactions, so a few risky merchants can hurt approval rates for everyone. Clip also fronts merchants their money quickly after a sale — which is valuable to legitimate merchants, but exactly what a fraudulent one would exploit. Because of this, Clip needs a reliable way to tell which merchants are safe (low fraud, high approval) and which are risky, and to use that assessment at multiple points: deciding whether to approve a transaction in the first place, and deciding how fast a merchant gets paid afterward.

Your task

1. Train a model. You're given a merchant-level dataset with an `is_suspended` field. Suspension isn't usable as a target variable in its raw form — you'll need to decide how to turn it into one, and justify your choice. There's no single correct answer here; this is part of what's being evaluated.

2. Score merchants and apply it to transactions. Once trained, use the model to produce a risk score per merchant, and join that score onto the separate transaction-level dataset provided (by merchant ID) so every transaction has a risk score attached.

3. Propose two sets of rules based on the score: - **Pre-authorization**: which transactions should be allowed through, and which should be declined. As a bonus: How can the score be used to bundle merchants' transactions into different segments so that reputation with issuers can be maintained or enhanced. - **Post-authorization**: among transactions that were approved,

which merchants should be paid out immediately, and which should be held for manual review.

Be explicit about your reasoning and trade-offs at each step — how you built the target, how you validated the model, and how you chose your rule thresholds.

Requirements

- **Public GitHub repo.** All work must be uploaded to a GitHub repository that's publicly accessible for review — code, analysis, and any supporting files.
- **AI tool use is allowed, but must be transparent.** You may use LLMs or agentic coding tools. If you do, document how — ideally in enough detail that someone else could reproduce your workflow (prompts used, what was generated vs. written by hand, etc.).
- **Separate model evaluation from business impact.** These are two different questions: "is the model good?" and "does using it actually help the business?" Each stage needs its own KPIs to support its decisions. Proposing, building, and justifying these KPIs is part of the challenge.